Math **163** Name (PLEASE PRINT) _____

Problem Set Score_____

Please complete the table below. You may submit only problems that you finish completely. Points will be taken off for any unfinished work submitted.

Problem	Points	Completed (Y)	Score
1	**2**		
2	**2**		
3	**2**		
4	**2**		
5	**2**		
6	**2**		
7	**2**		
8	**2**		
9	**2**		
10	**2**		
11	**2**		
12	**2**		
13	**2**		
14	**2**		
15	**2**		
		Total Score	

You are free to talk to anyone

Instructions

- All work must be done using a **MECHANICAL PENCIL following instructions given**.
- If you have any questions at all, please **ask me**. This includes the following possibilities:
    - something is **illegible,** or **ambiguous**, or **unclear** to you.
- **You may use your homework, notes, books, and calculators unless otherwise stated during the exam.**
- **You will not get any partial credit**. You have to show **absolutely all your work** on all questions, **exactly as I do on the board in class, with no changes, variations, or omissions in form, unless otherwise noted whether orally or in writing**. If your answer is correct, but your procedure is incorrect or is not allowed by the instructions, you will get no credit all. All work must be displayed using **correct syntax**, and all **explanations**, written in **complete sentences**, must be complete.
- **You must write neatly, clearly and legibly. If you run out of room, you may not turn the page, but you must continue the work on the facing page (on the left side) labelling your work appropriately. You must draw every straight-line segment other than those in the list: (**$=, \neq, \equiv, <, \leq, >, \geq, +, -, \times, \div, \ \neg, \vee, \wedge, \parallel, \perp, \backslash, \rightarrow, \leftrightarrow, \mapsto$**) with your ruler. At least 1 point will be taken off for each instance of violation. Signs of equality or equivalence must line up, one below the other on the left of expressions. Every symbol has a specific meaning and may be used only in this sense and in no other. Syntax defined in the notes must be followed exactly to the letter.**
- Remember that 'does not make sense' is a possible answer.
- I can grade you only on what you **actually** put on paper; I cannot read minds. You cannot say later that what you had written down is not what you had meant. **You need to learn to write exactly what you mean and mean exactly what you write**.
- **The answers for a problem should be noted on the page on which the problem appears in the areas indicated. All work for the problem should be shown on sheets that follow the page on which the problem appears numbered as follows. Work for a problem that occurs on page n should be on blank (not ruled) sheets after page n, numbered, n0, n1, n2 etc. on the lower right-hand corner as in this document. For example, work for a problem on page 502 should be on pages numbered 5020, 5021, 5022 etc. as needed. Work that corresponds to different parts must be separated from each other by striaght lines drawn with a ruler parallel to the edges of the paper and labelled correctly as in the Pages from Student Work as found under Files on CANVAS.**
- **You should submit a single document.**
- **As shown in the class means as shown in the notes.**

You are free to talk to anyone

D Define Y:= Yes N:=No

D Define Pf:= Proof W:=Witness

R **Note that in a set an element may occur exactly once**. Therefore there may not be any repeated elements in either *A* or *B* in the above. **The sets *A* and *B* will occur in several problems on this test**. **So, if you do not have the correct sets, all your answers will be wrong.**

R If answers to different instances of some question on the exam contradict each other you will get a score of zero for every instance.

R **You will lose points for making any extraneous mark on the exam.**

R You may only submit complete answers to questions marked explicitly on the exam. You will lose points for unfinished work. **Do not write anything down on any page if you cannot completely solve the problem or some part thereof that is indexed by a roman numeral.**

R Separate pieces of work for any part of any problem recorded on a blank page intended for the purpose must be indexed by the index of the problem and also by numbers starting at 0 that record the order in which the pieces were completed; such parts must be separated from each other by straight lines parallel to the edges of the paper drawn with a ruler. You will lose points otherwise.

R Any sloppiness, untidiness, and any departure from proper format (as indicated in class) will lead to a score of 0.

R **In arrow-diagrams, put the numbers in rectangles rather than ovals and draw the rectangles using rulers.**

2 (1) Find (and recall) a recursive definition for each of the functions below

$$\varphi \coloneqq n \mapsto n! : \mathbb{N} \longrightarrow \mathbb{N} \quad \text{and} \quad \psi \coloneqq n \mapsto 2^n : \mathbb{N} \longrightarrow \mathbb{N}$$

Prove by induction that: $\forall n \in \mathbb{N} \;\; \varphi(n+4) \geq \psi(n+4)$

D φ:

BC

$\varphi(0) \coloneqq 1$

RcS

$\forall n \in \mathbb{N} \;\; \varphi(n+1) \coloneqq (n+1)\varphi(n)$

D ψ:

BC

$\psi(0) \coloneqq 1$

RcS

$\forall n \in \mathbb{N} \;\; \psi(n+1) \coloneqq 2\,\psi(n)$

1.0

$P(n) \coloneqq (\varphi(n{+}4) \geq \psi(n{+}4))$

SPf

BC

$$\begin{array}{lll} & \varphi(4) & \psi(4) \\ = & 4! & = 2^4 \\ = & 24 & = 16 \end{array}$$

$$\frac{24 \geq 16}{P(0)}$$

RcS

$$\frac{P(n)}{\begin{array}{ll} & \varphi(n{+}5) \\ = & (n{+}5)\varphi(n{+}4) \\ \geq & (n{+}5)\psi(n{+}4) \\ \geq & 2\psi(n{+}4) \\ = & \psi(n{+}5) \end{array}}{P(n{+}1)}$$

EPf

2 (2) Starting with sets A, B, C, D, E and F with $a \coloneqq \nu(A)$, $b \coloneqq \nu(B)$, $c \coloneqq \nu(C)$, $d \coloneqq \nu(D)$, $e \coloneqq \nu(E)$ and $f \coloneqq \nu(F)$, use as many of the operations in the list below as many times as you wish:

$$\left\{ (\ )^c,\ \cup,\ \cap,\ \setminus,\ \prod,\ \coprod,\ Fnc(A,B),\ (\ ),\ \sim \right\}$$

(i) to construct a set S whose cardinality appears below, and

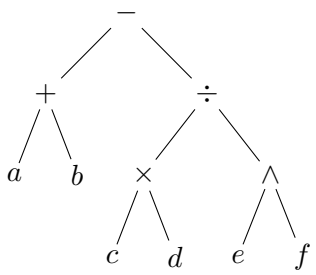
$$\nu(S) = a + b - c \times d \div e \wedge f$$

(ii) to construct word-problems whose solutions require counting the number of elements of the sets you constructed in (i). Note that the numbers are such that $\nu(S) \in \mathbb{N}$. Draw a formation-tree for the expression on the facing side. Draw a formation-tree for the expression that you have constructed. Prove that the cardinality of the set that you have constructed equals the given expression. Write the word problem step by step respecting the formation-tree of the expression that you have constructed.

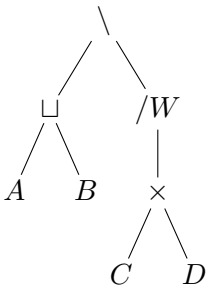
$a = 5,\ b = 4,\ c = 2,\ d = 3,\ e = 2,\ f = 1.$

$G \coloneqq Fnc(F, E)$; $P \coloneqq C \times D$; W: eq. rel. on P, classes of size $\nu(G)$; $Q \coloneqq P/W$; $S \coloneqq (A \sqcup B) \setminus Q$, $Q \subseteq A \sqcup B$.

2.0



2.1



2.2

$$\frac{Q \subseteq A \sqcup B}{(A \sqcup B) = \big((A \sqcup B) \setminus Q\big) \cup Q}$$

$\nu(A \sqcup B) = \nu(A) + \nu(B) = a + b = \nu(S) + \nu(Q)$

$\nu(P) = \nu(C \times D) = \nu(C) \times \nu(D) = c \times d$

$\nu(Fnc(F, E)) = \nu(E)^{\nu(F)} = e \wedge f = \nu(G)$

$\nu(Q) = \nu(P/W) = \nu(P) \div \nu(G) = (c \times d) \div (e \wedge f)$

$\nu(S) = \nu(A \sqcup B) - \nu(Q) = (a + b) - \big((c \times d) \div (e \wedge f)\big) = 6$ ■

2.3

C: 2 breads. D: 3 fillings. $\rightarrow \nu(C \times D) = 6$ sandwiches.

E: 2 sizes. F: 1 cup. $\rightarrow \nu(Fnc(F, E)) = 2$ size-assignments.

Group the 6 sandwiches into groups of 2. $\rightarrow \nu(Q) = 3$ groups.

A: 5 people, table 1. B: 4 people, table 2. $\rightarrow \nu(A \sqcup B) = 9$.

One person leaves per group (3 groups). $\rightarrow \nu(S) = 9 - 3 = 6$ people remain.

2 (3) Define a relation on $\mathbb{N}$ as follows:

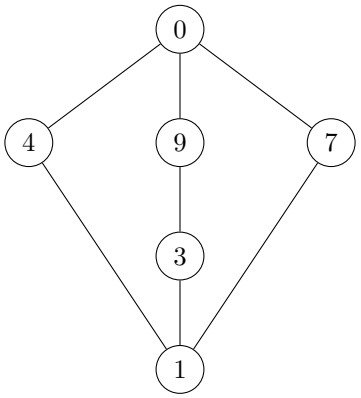
$$\forall x, y \in \mathbb{N} \quad (xRy) \coloneqq (x \mid y)$$

Prove that: $PO(R, \mathbb{N})$

Draw the Hasse Diagram for the $\{J_k \mid k \in 1..8\}$ under divisibility ordering $|$, that is:

$$HsDgrm(\{J_k \mid k \in 1..8\}, |)$$

$J = 00499317 \implies \{J_k \mid k \in 1..8\} = \{0, 1, 3, 4, 7, 9\}$.



3.0

$a,b,c \in \mathbb{N}$

$\dfrac{a = 1a}{a|a}$

3.1

$\dfrac{\forall a \in \mathbb{N}\ (a|a)}{\text{Rflx}(|)}$

3.2

$\dfrac{a{=}0\ a|b}{b = c{\bullet}0 = 0 = a}$

3.3

$\dfrac{a{\neq}0\ a|b\ b|a}{b{=}ca\ a{=}db}$

3.4

$\dfrac{a = db = d(ca) = (dc)a}{dc = 1}$

3.5

$\dfrac{dc = 1}{d{=}1\ c{=}1}$

3.6

$\dfrac{d{=}1\ c{=}1}{b = ca = 1a = a}$

3.7

$\dfrac{\forall a,b \in \mathbb{N}((a|b\ b|a)/(a{=}b))}{\text{AntSym}(|)}$

3.8

$\dfrac{a|b\ b|c}{b{=}c'a\ c{=}bd'}$

3.9

$\dfrac{c = bd' = (c'a)d' = (c'd')a}{a|c}$

3.10

$$\frac{\forall a,b,c\in\mathbb{N}((a|b\ b|c)/(a|c))}{\mathrm{Trns}(|)}$$

3.11

$$\frac{\mathrm{Rflx}(R)\ \mathrm{AntSym}(R)\ \mathrm{Trns}(R)}{\mathrm{PO}(R,\mathbb{N})}$$

2 (4) Define a relation on $\mathbb{Z}$ as follows:

$$\forall x, y \in \mathbb{Z} \quad (xRy) \coloneqq (x^2 = y^2)$$

Prove that: $Eqv(R, \mathbb{Z})$ and find the equivalence classes.

$\mathbb{Z}/R = \big\{\{0\}\big\} \cup \big\{\{-n, n\} \mid n \in \mathbb{N},\, n \geq 1\big\}$

4.0

$x,y,z \in \mathbb{Z}$

$\dfrac{x^2 = x^2}{xRx}$

4.1

$\dfrac{\forall x \in \mathbb{Z}\ (xRx)}{\mathrm{Rflx}(R)}$

4.2

$\dfrac{xRy}{x^2 = y^2}$

4.3

$\dfrac{x^2 = y^2}{y^2 = x^2}$

4.4

$\dfrac{y^2 = x^2}{yRx}$

4.5

$\dfrac{\forall x,y \in \mathbb{Z}\ (xRy\ /\ yRx)}{\mathrm{Sym}(R)}$

4.6

$\dfrac{xRy\ yRz}{x^2{=}y^2\ y^2{=}z^2}$

4.7

$\dfrac{x^2 = y^2 = z^2}{x^2 = z^2}$

4.8

$\dfrac{x^2 = z^2}{xRz}$

4.9

$\dfrac{\forall x,y,z \in \mathbb{Z}((xRy\ yRz)/(xRz))}{\mathrm{Trns}(R)}$

4.10

$$\frac{\mathrm{Rflx}(R)\ \mathrm{Sym}(R)\ \mathrm{Trns}(R)}{\mathrm{Eqv}(\mathbb{Z},R)}$$

4.11

$\mathrm{EqvCl}(x,R) = \{t \in \mathbb{Z} \mid t^2 = x^2\} = \{x,-x\}$

$x=0$: $\mathrm{EqvCl}(0,R)=\{0\}$ $x\neq 0$: $\mathrm{EqvCl}(x,R)=\{x,-x\}=\mathrm{EqvCl}(-x,R)$

$\mathbb{Z}/R = \{\{0\}\} \cup \{\{-n,n\} \mid n \in \mathbb{N}, n \geq 1\}$

2 (5) Define a relation on $\mathbb{N} \times \mathbb{N}$ as follows:

$$\forall (a,b),(c,d) \in \mathbb{N} \times \mathbb{N} \quad \big((a,b)R(c,d)\big) \coloneqq \big(a+b=c+d\big)$$

Prove that: $Eqv(R, \mathbb{N} \times \mathbb{N})$ and find the equivalence classes.

$(\mathbb{N} \times \mathbb{N})/R = \{\{(0,k),(1,k-1),\ldots,(k,0)\} \mid k \in \mathbb{N}\}$

5.0

$(a,b),(c,d),(e,f) \in \mathbb{N}\times\mathbb{N}$

$$\frac{a+b = a+b}{(a,b)R(a,b)}$$

5.1

$$\frac{\forall (a,b)\in\mathbb{N}\times\mathbb{N}\ ((a,b)R(a,b))}{\mathrm{Rflx}(R)}$$

5.2

$$\frac{(a,b)R(c,d)}{a+b = c+d}$$

5.3

$$\frac{a+b = c+d}{c+d = a+b}$$

5.4

$$\frac{c+d = a+b}{(c,d)R(a,b)}$$

5.5

$$\frac{\forall (a,b),(c,d)\in\mathbb{N}\times\mathbb{N}((a,b)R(c,d)/(c,d)R(a,b))}{\mathrm{Sym}(R)}$$

5.6

$$\frac{(a,b)R(c,d)\ (c,d)R(e,f)}{a+b=c+d\ c+d=e+f}$$

5.7

$$\frac{a+b = c+d = e+f}{a+b = e+f}$$

5.8

$$\frac{a+b = e+f}{(a,b)R(e,f)}$$

5.9

$$\frac{\forall (a,b),(c,d),(e,f)\in\mathbb{N}\times\mathbb{N}(((a,b)R(c,d)\ (c,d)R(e,f))/(a,b)R(e,f))}{\mathrm{Trns}(R)}$$

5.10

$$\frac{\mathrm{Rflx}(R)\ \mathrm{Sym}(R)\ \mathrm{Trns}(R)}{\mathrm{Eqv}(\mathbb{N}\times\mathbb{N},R)}$$

5.11

$k \coloneqq a+b.\ \mathrm{EqvCl}((a,b),R) = \{(x,y)\in\mathbb{N}\times\mathbb{N} \mid x+y=k\} = \{(0,k),(1,k-1),\ldots,(k,0)\}$, *size* $k+1$

$(\mathbb{N}\times\mathbb{N})/R = \{\{(0,k),\ldots,(k,0)\} \mid k\in\mathbb{N}\}$

2 (6) Define a relation on $\mathbb{N} \times \mathbb{N}$ as follows:

$$\forall (a,b),(c,d) \in \mathbb{N} \times \mathbb{N} \quad \big((a,b)R(c,d)\big) \coloneqq \big(a^2 + b^2 = c^2 + d^2\big)$$

Prove that: $Eqv(R, \mathbb{N} \times \mathbb{N})$ and find the equivalence classes.

$(\mathbb{N} \times \mathbb{N})/R = \big\{\{(x,y) \in \mathbb{N} \times \mathbb{N} \mid x^2 + y^2 = n\} \;\big|\; n \in \mathbb{N},\ \exists (x,y)\,(n = x^2 + y^2)\big\}$

6.0

$(a,b),(c,d),(e,f) \in \mathbb{N}\times\mathbb{N}$

$a^2+b^2 = a^2+b^2$

$(a,b)R(a,b)$

6.1

$\forall(a,b)\in\mathbb{N}\times\mathbb{N}\ ((a,b)R(a,b))$

Rflx(R)

6.2

$(a,b)R(c,d)$

$a^2+b^2 = c^2+d^2$

6.3

$a^2+b^2 = c^2+d^2$

$c^2+d^2 = a^2+b^2$

6.4

$c^2+d^2 = a^2+b^2$

$(c,d)R(a,b)$

6.5

$\forall(a,b),(c,d)\in\mathbb{N}\times\mathbb{N}((a,b)R(c,d)/(c,d)R(a,b))$

Sym(R)

6.6

$(a,b)R(c,d)\ (c,d)R(e,f)$

$a^2+b^2=c^2+d^2\ c^2+d^2=e^2+f^2$

6.7

$a^2+b^2 = c^2+d^2 = e^2+f^2$

$a^2+b^2 = e^2+f^2$

6.8

$a^2+b^2 = e^2+f^2$

$(a,b)R(e,f)$

6.9

$\forall(a,b),(c,d),(e,f)\in\mathbb{N}\times\mathbb{N}(((a,b)R(c,d)\ (c,d)R(e,f))/(a,b)R(e,f))$

Trns(R)

6.10

$$\frac{\mathrm{Rflx}(R)\ \mathrm{Sym}(R)\ \mathrm{Trns}(R)}{\mathrm{Eqv}(\mathbb{N}\times\mathbb{N},R)}$$

6.11

$n \coloneqq a^2+b^2.\ \mathrm{EqvCl}((a,b),R) = \{(x,y)\in\mathbb{N}\times\mathbb{N} \mid x^2+y^2=n\}$

$(\mathbb{N}\times\mathbb{N})/R = \{\{(x,y) \mid x^2+y^2=n\} \mid n\in\mathbb{N},\ \exists(x,y)(n=x^2+y^2)\}$

2 (7) Find a formula for the Fibonacci Numbers by solving the recurrence below. The Fibonacci numbers are defined recursively as follows:

$$\begin{cases} \rho(0) \coloneqq 0 \\ \rho(1) \coloneqq 1 \\ \forall n \in \mathbb{N} \quad \rho(n+2) - \rho(n+1) - \rho(n) = 0 \end{cases}$$

$c_2 = 1,\ c_1 = -1,\ c_0 = -1,\ R_0 = 0,\ R_1 = 1, \quad r \coloneqq \dfrac{-c_1}{2c_2} = \dfrac{1}{2}, \quad s \coloneqq \dfrac{\sqrt{c_1^2 - 4c_2c_0}}{2c_2} = \dfrac{\sqrt{5}}{2}$

Formula 11 (Induction and Recursion Notes p. 26).

$$\rho(n) = \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right)$$

7.0

$$\frac{\rho(n) = [(R_0(r+s)-R_1)/2s](r-s)^n + [(R_1-R_0(r-s))/2s](r+s)^n}{\rho(n) = (-1/\sqrt{5})((1-\sqrt{5})/2)^n + (1/\sqrt{5})((1+\sqrt{5})/2)^n} \left\langle \begin{matrix} R_0 \longleftarrow 0 \\ R_1 \longleftarrow 1 \\ r \longleftarrow 1/2 \\ s \longleftarrow \sqrt{5}/2 \end{matrix} \right\rangle$$

7.1

$r-s = (1-\sqrt{5})/2,\ r+s = (1+\sqrt{5})/2$

7.2

$\rho(n) = (1/\sqrt{5})[((1+\sqrt{5})/2)^n - ((1-\sqrt{5})/2)^n]$

7.3

n=0: $(1/\sqrt{5})(1-1) = 0 = \rho(0)$ n=1: $(1/\sqrt{5})\bullet\sqrt{5} = 1 = \rho(1)$

2 (8) Prove by induction that:

$$S(2,n) \coloneqq \sum_{k=0}^{n} k^2 \;=\; \frac{n(n+1)(2n+1)}{6}$$

8.0

$P(n) \coloneqq (S(2,n) = n(n+1)(2n+1)/6)$

SPf

BC

$S(2,0) = 0^2 = 0$

$P(0)$

RcS

$P(n)$

$$\begin{aligned} & S(2,n+1) \\ = \; & n(n+1)(2n+1)/6 + (n+1)^2 \\ = \; & (n+1)\bullet[n(2n+1)+6(n+1)]/6 \\ = \; & (n+1)\bullet(2n^2+7n+6)/6 \\ = \; & (n+1)(n+2)(2n+3)/6 \end{aligned}$$

$P(n+1)$

EPf

2 (9) Use the method of differences shown in class to find a recursive definition of:

$$S(3,n) \coloneqq \sum_{k=0}^{n} k^3$$

Prove by induction that:

$$S(3,n) \coloneqq \sum_{k=0}^{n} k^3 = \big(S(1,n)\big)^2$$

BC

$S(3,0) \coloneqq 0$

RcS

$\forall n \in \mathbb{N} \;\; S(3,n+1) \coloneqq S(3,n) + (n+1)^3$

$S(1,n) = \dfrac{n(n+1)}{2}$.

9.0

$S(1,n) = n(n+1)/2$. $P(n) := (S(3,n) = (S(1,n))^2)$

SPf

BC

$S(3,0) = 0 = (S(1,0))^2$

$P(0)$

RcS

$P(n)$

$$
\begin{aligned}
& S(3,n+1) \\
= \ & (n(n+1)/2)^2 + (n+1)^3 \\
= \ & (n+1)^2 \bullet [n^2/4 + (n+1)] \\
= \ & (n+1)^2 \bullet (n+2)^2/4 \\
= \ & ((n+1)(n+2)/2)^2 \\
= \ & (S(1,n+1))^2
\end{aligned}
$$

$P(n+1)$

EPf

2 (10) Find (and recall) a recursive definition for each of the functions below

$$\beta \coloneqq n \mapsto n! : \mathbb{N} \longrightarrow \mathbb{N} \quad \text{and} \quad \gamma \coloneqq n \mapsto n^n : \mathbb{N} \longrightarrow \mathbb{N}$$

Prove by induction that: $\forall n \in \mathbb{N} \;\; \gamma(n) \geq \beta(n)$

D β:

BC

$\beta(0) \coloneqq 1$

RcS

$\forall n \in \mathbb{N} \;\; \beta(n+1) \coloneqq (n+1)\beta(n)$

D $\exp \coloneqq (a, n) \mapsto a^n$:

BC

$\exp(a, 0) \coloneqq 1$

RcS

$\forall n \in \mathbb{N} \;\; \exp(a, n+1) \coloneqq a \cdot \exp(a, n)$

$\gamma(n) \coloneqq \exp(n, n)$

10.0

$P(n) \coloneqq (\gamma(n) \geq \beta(n))$

SPf

BC

$$\frac{\gamma(0) = 0^0 = 1 = 0! = \beta(0)}{P(0)}$$

RcS

$$\frac{P(n)}{\dfrac{\beta(n+1) = (n+1)\beta(n) \leq (n+1)\gamma(n)}{\dfrac{n \leq n+1}{\gamma(n) = n^n \leq (n+1)^n}}}$$

$$\begin{aligned} & (n+1)\gamma(n) \\ = \ & (n+1)(n+1)^n \\ = \ & (n+1)^{n+1} \\ = \ & \gamma(n+1) \end{aligned}$$

$$\frac{}{P(n+1)}$$

EPf

2 (11) Find truth-tables for each of 16 binary connectives $(c_k \mid k \in 1..(16))$ for Boolean Logic and express each formula pc_kq in words.

Rows $(p, q) = (0, 0), (0, 1), (1, 0), (1, 1)$.

11.0

c_1

p	q	c_1
0	0	0
0	1	0
1	0	0
1	1	0

always false

c_2

p	q	c_2
0	0	0
0	1	0
1	0	0
1	1	1

p and q

c_3

p	q	c_3
0	0	0
0	1	0
1	0	1
1	1	0

p and not q

c_4

p	q	c_4
0	0	0
0	1	0
1	0	1
1	1	1

p

c_5

p	q	c_5
0	0	0
0	1	1
1	0	0
1	1	0

q and not p

c_6

p	q	c_6
0	0	0
0	1	1
1	0	0
1	1	1

q

c_7

p	q	c_7
0	0	0
0	1	1
1	0	1
1	1	0

exactly one of p, q

c_8

p	q	c_8
0	0	0
0	1	1
1	0	1
1	1	1

p or q

c_9

p	q	c_9
0	0	1
0	1	0
1	0	0
1	1	0

neither p nor q

c_{10}

p	q	c_{10}
0	0	1
0	1	0
1	0	0
1	1	1

p iff q

c_{11}

p	q	c_{11}
0	0	1
0	1	0
1	0	1
1	1	0

not q

c_{12}

p	q	c_{12}
0	0	1
0	1	0
1	0	1
1	1	1

if q then p

11.1

c_{13}

p	q	c_{13}
0	0	1
0	1	1
1	0	0
1	1	0

not p

c_{14}

p	q	c_{14}
0	0	1
0	1	1
1	0	0
1	1	1

if p then q

c_{15}

p	q	c_{15}
0	0	1
0	1	1
1	0	1
1	1	0

not both p and q

c_{16}

p	q	c_{16}
0	0	1
0	1	1
1	0	1
1	1	1

always true

2 (12) Define that set of truth-values $TV \coloneqq \{-1, 0, 1\}$

(i) Define truth-tables for negation, disjunction, conjunction, and implication and justify your definitions on the basis of your interpretation of the truth-values.

(ii) How many types arise as analogues of the types: tautology, contradiction, and contingency, and how can they be interpreted?

(iii) Is implication definable in terms on negation and disjunction in your system?

$-1 \coloneqq$F, $0 \coloneqq$Unknown, $1 \coloneqq$T. $\neg p \coloneqq -p$. $p \wedge q \coloneqq \min(p, q)$. $p \vee q \coloneqq \max(p, q)$. $p \to q \coloneqq \max(-p, q)$.

(ii) $\boxed{4}$ types: constant -1, constant 0, constant 1, contingent (versus 3 for $\{0, 1\}$).

(iii) Yes, $p \to q \coloneqq \neg p \vee q$ by construction.

12.0

p	q	$p \wedge q$	$p \vee q$	$p \rightarrow q$
1	1	1	1	1
1	0	0	1	0
1	−1	−1	1	−1
0	1	0	1	1
0	0	0	0	0
0	−1	−1	0	0
−1	1	−1	1	1
−1	0	−1	0	1
−1	−1	−1	−1	1

p	$\neg p$
1	−1
0	0
−1	1

2 (13) Define a relation Q on the set of propositions Prp in Propositional Calculus as follows:

$$\forall p, q \in Prp \quad pQq \coloneqq (p \leftrightarrow q)$$

(i) Prove that Q is and equivalence relation on Prp, that is, $Eqv(Q)$. Define $\mathbb{F} \coloneqq Prp/Q$, the set of equivalence classes $EqvCls(Prp, Q)$.

(ii) Prove that: $\forall p, q, r, s \in Prp \left(\dfrac{pQq \quad p \rightarrow r \quad rQs}{q \rightarrow s} \right)$

(iii) Define, in view of (ii) a relation $\mathcal{E}$ on $\mathbb{F}$ as follows: $\forall F, G \in \mathbb{F} \quad F\mathcal{E}G \coloneqq (F \Rrightarrow G)$, noting that $\Rrightarrow$ means that every proposition in the equivalence class F implies (in the ordinary sense $\rightarrow$ of PC) every proposition in the equivalence class G.

(iv) Prove that $PO(\mathcal{E}, \mathbb{F})$, that is, $\mathcal{E}$ is a partial order on $\mathbb{F}$.

13.0

$p,q,r \in \mathrm{Prp}$

$p \leftrightarrow p$ $(tautology)$

pQp

13.1

$\forall p \in \mathrm{Prp}\ (pQp)$

$\mathrm{Rflx}(Q)$

13.2

pQq

$p \leftrightarrow q$

13.3

$p \leftrightarrow q$

$q \leftrightarrow p$

13.4

$q \leftrightarrow p$

qQp

13.5

$\forall p,q \in \mathrm{Prp}\ (pQq/qQp)$

$\mathrm{Sym}(Q)$

13.6

$pQq\ qQr$

$p \leftrightarrow q\ q \leftrightarrow r$

13.7

$p \leftrightarrow q \leftrightarrow r$

$p \leftrightarrow r$

13.8

$p \leftrightarrow r$

pQr

13.9

$\forall p,q,r \in \mathrm{Prp}((pQq\ qQr)/pQr)$

$\mathrm{Trns}(Q)$

13.10

$$\frac{\mathrm{Rflx}(Q)\ \mathrm{Sym}(Q)\ \mathrm{Trns}(Q)}{\mathrm{Eqv}(Q)}$$

13.11

$\mathbb{F} \coloneqq \mathrm{Prp}/Q \coloneqq \{\mathrm{EqvCl}(p,Q) \mid p \in \mathrm{Prp}\}$

13.12

pQq, p→r, rQs given.

$$\frac{pQq}{p \leftrightarrow q}$$

13.13

$$\frac{p \leftrightarrow q}{q \rightarrow p}$$

13.14

$$\frac{rQs}{r \leftrightarrow s}$$

13.15

$$\frac{r \leftrightarrow s}{r \rightarrow s}$$

13.16

$$\frac{q \rightarrow p\ \ p \rightarrow r\ \ r \rightarrow s}{q \rightarrow s}$$

13.17

$F\mathcal{E}G \coloneqq (F \Longrightarrow G)$: *well-defined by* (ii) (*independent of representatives* $p{\in}F,r{\in}G$)

13.18

$$\frac{F,G,H \in \mathbb{F} \quad p,p' \in F}{pQp'}$$

13.19

$$\frac{pQp'}{p \leftrightarrow p'}$$

13.20

$$\frac{p \leftrightarrow p'}{p \rightarrow p'}$$

13.21

$$\frac{\forall F \in \mathbb{F}\ (F\mathcal{E}F)}{\mathrm{Rflx}(\mathcal{E})}$$

13.22

$$\frac{F\mathcal{E}G\ G\mathcal{E}F}{p_0 \rightarrow q_0\ q_0 \rightarrow p_0}$$

13.23

$$\frac{p_0 \rightarrow q_0\ q_0 \rightarrow p_0}{p_0 \leftrightarrow q_0}$$

13.24

$$\frac{p_0 \leftrightarrow q_0}{p_0 Q q_0}$$

13.25

$$\frac{p_0 Q q_0}{F = G}$$

13.26

$$\frac{\forall F,G \in \mathbb{F}((F\mathcal{E}G\ G\mathcal{E}F)/(F=G))}{\mathrm{AntSym}(\mathcal{E})}$$

13.27

$$\frac{F\mathcal{E}G\ G\mathcal{E}H}{\forall p{\in}F,q{\in}G(p{\to}q)\ \forall q{\in}G,r{\in}H(q{\to}r)}$$

13.28

$$\frac{\forall p{\in}F,q{\in}G(p{\to}q)\ \forall q{\in}G,r{\in}H(q{\to}r)}{\forall p{\in}F,r{\in}H(p{\to}r)}$$

13.29

$$\frac{\forall p{\in}F,r{\in}H(p{\to}r)}{F\mathcal{E}H}$$

13.30

$$\frac{\forall F,G,H{\in}\mathbb{F}((F\mathcal{E}G\ G\mathcal{E}H)/F\mathcal{E}H)}{\mathrm{Trns}(\mathcal{E})}$$

13.31

$$\frac{\mathrm{Rflx}(\mathcal{E})\ \mathrm{AntSym}(\mathcal{E})\ \mathrm{Trns}(\mathcal{E})}{\mathrm{PO}(\mathcal{E},\mathbb{F})}$$

2 (14) Consider the functions:

$$\alpha \coloneqq n \mapsto (n^n)! : \mathbb{N} \longrightarrow \mathbb{N}$$
$$\beta \coloneqq n \mapsto (n!)^n : \mathbb{N} \longrightarrow \mathbb{N}$$
$$\gamma \coloneqq n \mapsto (n!)! : \mathbb{N} \longrightarrow \mathbb{N}$$
$$\delta \coloneqq n \mapsto (n)^{n!} : \mathbb{N} \longrightarrow \mathbb{N}$$

Formulate a conjecture about the eventual behaviour of these functions with respect to value-wise ordering and prove your conjecture by induction.

Conjecture: $\forall n \in \mathbb{N}\,(n \geq 4) \quad \beta(n) < \delta(n) < \gamma(n) < \alpha(n)$

14.0

n=3: β=216 δ=729 γ=720 α=27! (γ,δ *swapped*; *close*: 720 *vs* 729)
n=4: β=331,776 δ=4^{24}≈2.81×10^{14} γ=24!≈6.20×10^{23} α=256!
n=5: β≈2.49×10^{10} δ=5^{120}≈7.5×10^{83} γ=120!≈6.7×10^{198} α=3125!

14.1

$$\frac{331{,}776 < 4^{24} < 24! < 256!}{P(4)}$$

14.2

β<δ (*base* n=4; *other two below*): $a(n)$:=ln $\beta(n)$=n•ln(n!), $b(n)$:=ln $\delta(n)$=n!•ln n

$$\frac{a(n)<b(n)}{(n+1)\ln(n!)+(n+1)\ln(n+1) < (n+1)[n!/n\bullet\ln n+\ln(n+1)]}$$

14.3

$$\frac{n!/n \le n!-1}{a(n+1) < b(n+1)}$$

14.4

$n \le n!(n-1)$, *true* $\forall n \ge 4$ *since* $n!(n-1) \ge 24\bullet 3 = 72 \gg n$

14.5

$\delta<\gamma$: $\ln n<\ln(n!)$ $(n\geq 2)$

$\ln n < \ln(n!)$ $(n\geq 2)$

$n!\bullet\ln n < n!\bullet\ln(n!) \leq \ln((n!)!) = \ln \gamma(n)$

14.6

$\gamma<\alpha$: $n! \leq n^n$ $(n\geq 2)$, *each of* $n!$ *factors is* $\leq n$

$n! \leq n^n$ $(n\geq 2)$

$\alpha(n) = (n^n)! \geq (n!)! = \gamma(n)$

14.7

$P(4)$ 14.2 14.3 14.5 14.6

$\forall n\in\mathbb{N}(n\geq 4)$ $(\beta(n)<\delta(n)<\gamma(n)<\alpha(n))$

2 (15) Use the method of differences shown in class to find a recursive definition for:

$$S(p,n) \coloneqq \sum_{k=0}^{n} k^p \quad \text{where } p,n \in \mathbb{N}$$

Using your recursive definition, find a formula for $S(p,n)$ and prove by induction that your formula is valid.

BC

$S(p,0) \coloneqq 0^p$

RcS

$\forall n \in \mathbb{N} \;\; S(p,n+1) \coloneqq S(p,n) + (n+1)^p$

$$S(p,n) = \frac{1}{p+1}\left((n+1)^{p+1} - \sum_{j=0}^{p-1}\binom{p+1}{j}S(j,n)\right), \qquad S(0,n) = n+1$$

15.0

$Q(n) \coloneqq (\Sigma(k{=}0..n)[(k{+}1)\hat{}(p{+}1){-}k\hat{}(p{+}1)] = (n{+}1)\hat{}(p{+}1))$

SPf

BC

$1\hat{}(p{+}1) - 0\hat{}(p{+}1)$

$= 1$

$= (0{+}1)\hat{}(p{+}1)$

$Q(0)$

RcS

$Q(n)$

$\Sigma(k{=}0..n{+}1)[(k{+}1)\hat{}(p{+}1){-}k\hat{}(p{+}1)]$

$= \Sigma(k{=}0..n)[(k{+}1)\hat{}(p{+}1){-}k\hat{}(p{+}1)] + [(n{+}2)\hat{}(p{+}1){-}(n{+}1)\hat{}(p{+}1)]$

$= (n{+}1)\hat{}(p{+}1) + (n{+}2)\hat{}(p{+}1) - (n{+}1)\hat{}(p{+}1)$

$= (n{+}2)\hat{}(p{+}1)$

$= ((n{+}1){+}1)\hat{}(p{+}1)$

$Q(n{+}1)$

EPf

15.1

$(k{+}1)\hat{}(p{+}1) - k\hat{}(p{+}1) = \Sigma(j{=}0..p)\ C(p{+}1,j)\ k\hat{}j$

$(k{+}1)\hat{}(p{+}1) - k\hat{}(p{+}1) = \Sigma(j{=}0..p)\ C(p{+}1,j)\ k\hat{}j \qquad Q(n)$

$\Sigma(j{=}0..p)\ C(p{+}1,j)\ S(j,n) = (n{+}1)\hat{}(p{+}1)$

15.2

$S(p,n) = 1/(p{+}1) \bullet [(n{+}1)\hat{}(p{+}1) - \Sigma(j{=}0..p{-}1)\ C(p{+}1,j)\ S(j,n)]$

15.3

$p{=}2$: $S(2,n)$

$= \frac{1}{3}[(n{+}1)^3{-}(n{+}1){-}3{\bullet}n(n{+}1)/2]$

$= n(n{+}1)(2n{+}1)/6$ ✓ (*matches* 2(8))